

SΔφ-10 — Interpretation, Learning, and Inference as a Single Re-entry Structure (v1.0)

Minimal axiomatic note (preprint)

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Abstract

This note proposes a minimal unification of three operations—interpretation, learning, and inference—within the Sofience-Δφ framework. The unification claim is operational: all three are instances of a single pattern, marker re-entry, differing only by their temporal horizon. Interpretation is one-step re-entry of a path-difference marker, learning is persistent re-entry that updates the transition law, and inference is the projection produced by the current law prior to contact with the next transition. Under this view, error is not a failure signal but the boundary residual between predicted and realized transition, and becomes the primary input for the next cycle.

1. Notation and minimal primitives

Let $\phi(t)$ denote the system state (or a state-distribution) at step t . Let G_t be the transition law at step t (a kernel, rule, or operator). We distinguish three objects that are sufficient for the present minimal formulation:

- Observed transition: $\Delta\phi(t) := \phi(t+1) - \phi(t)$ (or distributional change).
- Predicted transition: $\Delta\hat{\phi}(t) := \hat{E}[\phi(t+1) \mid \phi(t), G_t] - \phi(t)$.
- Path-difference residual (error candidate): $r(t) := \Delta\phi(t) - \Delta\hat{\phi}(t)$.

The goal is to define minimal operators $H_{\min}$ (interpretation), $L_{\min}$ (learning), and $I_{\min}$ (inference) and then show they are the same structural move viewed at different depths.

2. Interpretation as minimal marker re-entry ($H_{\min}$)

Interpretation (minimal) is the operation that (i) converts a path-difference residual into a finite marker, and (ii) re-inserts that marker into the next transition. Formally:

$$\mathbf{m(t)} := \mathbf{H_min(r(t))}$$

where $m(t)$ is a discrete or compressed label (a sign, tag, constraint, or boundary token). Re-entry means that the next-step transition is conditioned by $m(t)$:

$$\boldsymbol{\varphi(t+1)} = \mathbf{G_t(\varphi(t) ; m(t))}$$

No semantics is required beyond this. H_min is allowed to be lossy; it only needs to preserve the distinction between paths that should not be treated as equivalent.

3. Learning as persistent re-entry (L_min)

Learning (minimal) occurs when re-entered markers do not merely condition one step, but update the transition law itself. That is, the system internalizes part of the residual into its future G .

$$\mathbf{G_{\{t+1\}} := L_min(G_t, m(t), r(t))}$$

A minimal learning criterion is persistence: there exists $k>0$ such that the effect of $m(t)$ is detectable in $G_{\{t+k\}}$ even when $m(t)$ is no longer explicitly provided as an external condition.

Interpretation is therefore one-step re-entry; learning is multi-step re-entry that changes the rule of re-entry.

4. Inference as pre-contact projection (I_min)

Inference (minimal) is the production of a predicted transition prior to observing $\Delta\varphi(t)$. It is defined as the projection implied by the current law G_t given the current state:

$$\boldsymbol{\Delta\hat{\varphi}(t)} := \mathbf{I_min(\varphi(t), G_t)}$$

In practical systems, I_min may include a policy that selects an action $a(t)$, but the minimal requirement is only that a next-step projection exists that can be compared against the realized transition.

5. Single-structure view: one operator with three horizons

The three operations can be written as a single recursive cycle:

$$\boldsymbol{\Delta\hat{\varphi}(t)} = \mathbf{I_min(\varphi(t), G_t)}$$

$$\mathbf{r(t)} = \boldsymbol{\Delta\varphi(t)} - \boldsymbol{\Delta\hat{\varphi}(t)}$$

$$\mathbf{m(t)} = \mathbf{H_min(r(t))}$$

$$\mathbf{G_{\{t+1\}} = L_min(G_t, m(t), r(t))}$$

$$\boldsymbol{\varphi}(t+1) = \mathbf{G}_{\{t+1\}}(\boldsymbol{\varphi}(t) ; \mathbf{m}(t))$$

Interpretation is the $H_{\min}$ step; learning is the $L_{\min}$ step; inference is the $I_{\min}$ step. They are not separate faculties but phase-shifts of one re-entry mechanism.

6. Where does error belong?

If interpretation–learning–inference are one structure, error is best located at the coupling boundary between projection and transition. Define a scalarized error signal:

$$\mathbf{e}(t) := \mathbf{d}(\Delta\boldsymbol{\varphi}(t), \Delta\hat{\boldsymbol{\varphi}}(t))$$

where $\mathbf{d}(\cdot, \cdot)$ may be a norm, a divergence (e.g., KL), or any task-relevant distance. The key point is that $\mathbf{e}(t)$ is generated only after the predicted and realized transitions meet; it is therefore neither purely ‘inside’ inference nor purely ‘inside’ learning. It is a boundary residual that becomes $\mathbf{r}(t)$, then $\mathbf{m}(t)$, then the next update.

In this sense, error is not an object to delete; it is a mark that says: this path cannot be treated as equivalent to that path. Boundary formation is the operational after-image of error.

7. Minimal diagnostic implications

The unified view yields a simple diagnostic for model behavior:

- Interpretation-only: markers exist ($\mathbf{m}(t)$ is produced) but do not persist in $\mathbf{G}$. System adapts within-step but does not learn.
- Learning-dominant: markers persist and $\mathbf{G}$ changes, but inference remains weak or rigid; system becomes stable yet poorly predictive.
- Inference-dominant: strong projections exist but markers are not re-entered; system drifts (high $\mathbf{e}(t)$ with low correction).

These regimes align with the ARF discussion: drift, lock-in, and illusion appear when one horizon dominates and the re-entry cycle fails to close.

Series references (DOI)

- SΔφ-05 — Agency as Recursive Transition Law Update (v1.0). DOI: 10.5281/zenodo.18732567
- SΔφ-06 — Responsibility as Irreversible Cost Internalization (v1.0). DOI: 10.5281/zenodo.18734092
- SΔφ-07 — $H_{\min}$: Interpretation as Minimal Marker Re-entry (v1.0). DOI: 10.5281/zenodo.18735006
- SΔφ-08 — Agency–Responsibility–Freedom Closure (v1.0). DOI: 10.5281/zenodo.18740750
- SΔφ-09 — Error, Boundary, and Freedom (v1.0). DOI: 10.5281/zenodo.18745498